

SageMath: One Last Tango with Cybersecurity



In the previous article in this series (published in the November 2025 issue of *OSFY*), we had a detailed discussion on secure key exchange and how the Diffie–Hellman key exchange scheme can be implemented using SageMath. As promised, in this article we will discuss digital signatures, hashing, and message authentication.

Let's begin by trying to have a theoretical understanding of the concepts of digital signatures, hashing and message authentication.

In the previous article, we looked at a practical situation where a digital signature becomes essential—when your friend emails you agreeing to lend ₹1 million and later denies ever sending the message. A digital signature is a mathematical technique used to verify the authenticity and integrity of digital messages or documents. When a

message carries a valid digital signature, the recipient can be confident that it was indeed sent by the claimed sender and has not been tampered with. An added advantage is that a properly generated digital signature is far more trustworthy than a handwritten signature, since it is backed by strong cryptographic guarantees. Digital signatures can be created using various cryptographic algorithms such as RSA, DSA, and ECDSA, each offering different security properties and performance characteristics.